MOSAIC: Calibrated Per-Cell Uncertainty for Unpaired Single-Cell Multi-Omics Integration in Disease Atlases

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Abstract

Disease atlases built from single-cell multi-omics measurements are now the de-facto substrate for precision-medicine hypothesis generation: which regulatory state drives a tumor subtype, which immune population expands in severe COVID-19, which enhancer is primed in a neurodevelopmental disorder. In practice, almost all such atlases are unpaired across modalities — the Human Cell Atlas transcriptome and the ENCODE chromatin-accessibility compendium describe overlapping populations through different molecular lenses, but no individual cell was measured in both. Every existing integration method returns a point-estimate cell-to-cell alignment with no signal to distinguish a trustworthy match from an artifact. The downstream consequence is clinically relevant: wrong alignments silently corrupt joint cluster discovery, regulatory inference, and spatial deconvolution — the steps that translate an atlas into a therapeutic hypothesis. We introduce MOSAIC, a method that turns unpaired multi-omics alignment into a well-defined probabilistic problem and reports a *cluster-resolved alignment entropy* as a per-cell uncertainty measure. MOSAIC combines per-modality information-bottleneck VAEs with a cluster-centroid cross-modal prediction target, an orthogonal-Procrustes post-alignment, and entropic Sinkhorn optimal transport whose transport plan is marginalized over partner cluster labels. The marginalization is the key technical insight: cell-level row entropy on the Sinkhorn plan *anti-correlates* with alignment error ($\rho = -0.65$) because of the geometry of dense clusters; cluster-resolved entropy removes this confound. On three paired benchmarks (PBMC 10x Multiome, E18 mouse brain 5k Multiome, and 10x CITE-seq), MOSAIC outperforms SCOT and uniPort baselines on every alignment metric and discriminates wrong-cluster assignments with AUROC 0.88–0.95. In ten simulated clinical-deployment scenarios — five immunodeficiencies on PBMC (CD8 lymphopenia, NK deficiency, B-cell aplasia, monocytopenia, Treg deficiency; mean AUROC 0.952) and five neurological diseases on Brain 5k (Alzheimer's, epilepsy, multiple sclerosis, ALS, PLX5622 microglia depletion; mean AUROC 0.993) — cluster-resolved entropy correctly flags the depleted cell populations with no disease-specific training. On CITE-seq, the same signal identifies PD-1$^+$TIGIT$^+$ exhausted T cells as having significantly higher transcriptome-proteome alignment uncertainty than fresh T cells ($p = 9.2 \times 10^{-5}$) — a clinically-relevant signature for checkpoint-inhibitor response. A cross-tissue negative control (PBMC RNA $\times$ mouse-brain ATAC, zero shared cell types) yields 4.2$\times$ higher mean entropy than within-dataset runs. MOSAIC provides the calibrated filtering substrate that clinical multi-omics integration currently lacks.

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1. Introduction

Modern translational research increasingly relies on single-cell multi-omics atlases as the backbone of disease characterization. The Human Cell Atlas (Regev et al., 2017), the Human Tumor Atlas Network (Rozenblatt-Rosen et al., 2020), the COVID-19 Cell Atlas (Stephenson et al., 2021), and the HuBMAP project (HuBMAP Consortium, 2019) all operate on the same premise: by cataloguing transcriptional, epigenomic, and proteomic states of individual cells in health and disease, we obtain the substrate from which biomarkers, drug targets, and mechanistic hypotheses are drawn. A central practical obstacle to this premise is that the relevant measurements are almost never taken on the same cell. Single-cell RNA-seq (scRNA-seq) and single-cell ATAC-seq (scATAC-seq) atlases profile overlapping cell populations through different molecular lenses, but each individual cell is destroyed at measurement; paired multi-omics assays (10x Multiome, CITE-seq) remain expensive and cover a small fraction of existing atlas data. Unpaired integration — mapping a cell measured in modality A to a corresponding cell measured in modality B — has therefore become a standard preprocessing step for joint clustering, regulatory inference, cell-communication analysis, and downstream disease-association studies (Demetci et al., 2022; Cao and Gao, 2022; Cao et al., 2022; Hao et al., 2024).

A clinically consequential gap pervades this literature: no method reports per-cell uncertainty about the match it produces. Existing approaches — Seurat v5 bridge integration (Hao et al., 2024), GLUE (Cao and Gao, 2022), SCOT (Demetci et al., 2022), uniPort (Cao et al., 2022), MultiVI (Ashuach et al., 2023), LIGER (Welch et al., 2019), Harmony (Korsunsky et al., 2019) — all return a point estimate. Some expose a nearest-neighbor rank; none tell a researcher whether a particular cell has been matched reliably. The downstream consequence is clinically important. Incorrect alignments propagate silently into joint cluster labels, differential-accessibility tests, enhancer–gene linkage inferences, and spatial deconvolution outputs — the steps that produce the therapeutic hypotheses these atlases are built to generate. A candidate regulatory program that is in fact an artifact of mismatched cells can redirect a pharmaceutical program by years; in a disease atlas, a cell population that exists only in the diseased tissue can be silently "matched" to whatever healthy population happens to be nearest in a shared embedding, producing a spurious null finding precisely where novelty is most clinically valuable.

We introduce MOSAIC (Multi-Omics Stochastic Alignment via Information-theoretic Coupling), a method that reframes unpaired single-cell multi-omics integration as a probabilistic problem and reports a cluster-resolved alignment entropy as a per-cell uncertainty measure. MOSAIC composes three components. Each modality is encoded by its own information-bottleneck variational autoencoder (IB-VAE) whose latent is constrained to both reconstruct the input and predict a cluster-centroid summary of the partner modality; the cluster-centroid target is a critical design choice, because naively using the partner cell's own feature vector causes catastrophic per-cell memorization while the cluster-centroid formulation forces the latent to retain cluster identity — a generalizing signal — rather than individual cell fingerprints. The two independently-trained IB-VAE latents live in different rotations of the same underlying space, so we fit an orthogonal Procrustes rotation on shared cluster centroids to bring them into a common frame; this post-hoc alignment is cheap, interpretable, and avoids the joint-optimization complexity of GLUE and MultiVI. Finally

we solve an entropic optimal-transport problem between the two latent clouds and, rather than reporting per-row entropy of the transport plan — which we show is miscalibrated as a per-cell uncertainty signal because of the geometry of dense clusters — we marginalize each row over the partner modality's cluster labels. The resulting cluster-level distribution has a natural entropy that captures cross-cluster uncertainty and predicts (AUROC 0.88–0.95) which cells have been assigned to the wrong cluster.

The core technical insight, phrased as a negative result, is that the obvious candidate uncertainty signal — row entropy of an entropic-OT transport plan — is not merely weak but directionally wrong. On PBMC 10k it yields Spearman $\rho = -0.65$ against true alignment error: cells in dense clusters have many valid within-cluster candidates (inflating entropy) but also a closer true partner (reducing error), so the two effects reverse the expected correlation. We believe this pathology generalizes to any OT-based integration method that exposes per-row entropy as a confidence score, and the marginalization cure is cheap and principled. On three paired benchmarks — 10x Multiome PBMC 10k (immune cells, 11,303 paired cells, 18 cell types), 10x Multiome E18 mouse brain 5k (neural development, 4,531 cells, 20 types), and 10x CITE-seq PBMC (RNA + 14 protein markers, 16 types) — MOSAIC outperforms SCOT, uniPort, and non-IB raw-feature baselines on every alignment metric, identifies wrong-cluster cells with AUROC 0.88–0.95, and — most importantly for clinical use — in a leave-one-cluster-out missing-type test (the computational analog of "this cell population exists in the disease atlas but not in the healthy reference") flags absent-type cells with mean AUROC 0.96 on PBMC, 0.995 on Brain, and 0.946 on CITE-seq. A cross-tissue negative control (PBMC RNA $\times$ mouse-brain ATAC, no shared cell types) shows 4.2$\times$ higher mean cluster entropy than within-dataset controls, confirming the signal correctly reports high uncertainty when nothing truly aligns.

Generalizable Insights about Machine Learning in the Context of Healthcare

- **Unpaired multi-omics integration is the modal setting for clinical atlases, not a methodological afterthought.** Paired assays are expensive; the data that actually populates disease atlases is overwhelmingly unpaired across modalities. A method that reports only a point-estimate alignment leaves a clinician or translational researcher unable to distinguish a trustworthy regulatory hypothesis from a silent artifact.

- **Calibrated per-cell uncertainty is the clinically actionable primitive.** An alignment with quantified uncertainty enables filtering (discard low-confidence matches before downstream analysis), triage (prioritize high-confidence matches for validation), and detection of disease-specific populations that exist in one atlas but not in the reference. Clinical utility is unlocked by the uncertainty signal, not the alignment itself.

- **Naive entropy on OT transport plans is miscalibrated in a characteristic way.** Row entropy of an entropic Sinkhorn plan anti-correlates with alignment error because dense clusters jointly inflate entropy and shrink the true-partner distance. The fix — marginalize over partner cluster labels before taking the entropy — is

cheap, principled, and, we conjecture, applicable to any OT-based integration method that wishes to expose per-cell confidence.

- **Cross-modal prediction targets must be cluster-level, not cell-level, to generalize.** In our development, switching from per-cell partner features to cluster-centroid targets produced a 3.6× improvement in label-transfer accuracy without any other change. Any architecture that couples two independently-encoded modalities through a cross-modal prediction head is susceptible to per-cell memorization unless the target has cluster-level structure.

- **One calibrated signal transfers across disease modalities without disease-specific training.** The same cluster-resolved entropy achieves AUROC 0.877–0.999 across ten simulated disease scenarios spanning immune (post-transplant, viral, autoimmune, post-rituximab, marrow-failure) and neurological (Alzheimer's, epilepsy, MS, ALS, PLX5622) cell-loss patterns, and simultaneously surfaces exhausted-T-cell and CD4/CD8 heterogeneity on CITE-seq — all with no scenario-specific label or threshold. This argues that a single MOSAIC deployment can serve as the quality-control substrate for a translational team working across indications.

2. Related Work

The unpaired multi-omics integration literature is crowded and largely driven by biological-benchmarking criteria: SCOT (Demetci et al., 2022) uses Gromov–Wasserstein optimal transport on intra-modality distance matrices; uniPort (Cao et al., 2022) uses coupled autoencoders with a domain-adversarial regularizer; MultiVI (Ashuach et al., 2023) jointly trains a shared-latent VAE with modality-specific decoders; GLUE (Cao and Gao, 2022) uses a graph-linked unified embedding constrained by prior biological graphs (e.g., peak-to-gene linkages); LIGER (Welch et al., 2019) and Harmony (Korsunsky et al., 2019) use non-negative matrix factorization and iterative clustering; and Seurat v5 bridge integration (Hao et al., 2024) re-uses a small paired dataset as an anchoring bridge between two unpaired datasets. None of these methods expose a per-cell probabilistic uncertainty on the match itself, and this is where MOSAIC is complementary: the IB-VAE + Procrustes + Sinkhorn scaffolding is standard (though the cluster-centroid cross-modal target we introduce is novel), and the main contribution is the cluster-resolved entropy as a calibrated uncertainty signal. Adjacent literature on uncertainty quantification in single-cell analysis — scANVI (Xu et al., 2021) exposes posterior variances on latent embeddings, uncertainty quantification has been applied to atlas-level cell-type transfer (Engelmann et al., 2022) — does not translate into cell-to-cell match confidence across modalities; no prior method, to our knowledge, reports per-cell uncertainty on an unpaired multi-omics match. Optimal transport for single-cell data is well-developed in temporal and spatial settings (Waddington-OT (Schiebinger et al., 2019), moscot (Klein et al., 2025), PASTE (Zeira et al., 2022), SCOT itself), and the row entropy of a Sinkhorn plan is sometimes used qualitatively as a fuzziness indicator; its directional miscalibration as a per-cell uncertainty measure (Spearman $\rho = -0.65$ against true error) has not been reported, and we view the marginalization cure as a portable contribution for this family. Clinical applications of single-cell atlases in tumor immune microenvironment characterization (Zheng et al., 2021), COVID-19 sever-

ity stratification (Stephenson et al., 2021), inflammatory bowel disease subtyping (Smillie et al., 2019), and autoimmune cell-population mapping (Zhang et al., 2021) each integrate unpaired datasets across modalities and tissues, and each is therefore susceptible to silent alignment errors that a calibrated uncertainty signal would catch.

3. Methods

MOSAIC takes two single-cell modalities A (e.g., RNA) and B (e.g., ATAC) measured on disjoint cell populations — or paired cells whose correspondence is treated as unknown — and produces a joint latent space in which paired cells map close together and a per-cell cluster-resolved alignment uncertainty score calibrated against true alignment error. The pipeline has three stages, laid out below as continuous prose because their interfaces are tightly coupled; the joint latent is shown in Figs. 1 and 2.

Each modality's encoder is a 3-layer MLP $d_x \to 512 \to 256 \to 64$ (with 2×64 output for $\mu, \log \sigma$ of the latent posterior), and the decoder mirrors the encoder; for RNA the decoder produces the three parameters of a zero-inflated negative binomial distribution over raw counts with cell-library-size scaling (reconstruction loss is the ZINB negative log-likelihood), for ATAC it produces per-peak Bernoulli logits against binarized peak accessibility (binary cross-entropy), and for CITE-seq proteins we use CLR-normalized values with a Gaussian likelihood. A cross-modal prediction head $g_\phi : z \to \mathbb{R}^{d_y}$ is attached to the latent; its target $y_{\text{cross}}(i)$ for cell i is the centroid, in the partner modality's 50-dim LSI/PCA space, of all cells that share i's cluster assignment. The full IB-VAE loss for modality m is

$$\mathcal{L}_m = \mathcal{L}_{\text{recon},m} + \beta \cdot \text{KL}(q(z|x) \,\|\, \mathcal{N}(0, I)) + \lambda_{\text{pred}} \cdot \|g_\phi(z) - y_{\text{cross}}\|_2^2, \tag{1}$$

with β annealed linearly over 30 epochs from 0 to its target and $\lambda_{\text{pred}} = 5$; training uses AdamW at lr 10^{-3} with a 10-epoch cosine warmup and 200 total epochs, and early stopping is disabled because the cross-modal loss is non-monotonic during the first half of training. We use $\beta = 0.001$ as the recommended default (an early $\beta = 0.01$ was defaulted based on a single-seed outlier and is retained in tables for traceability). The cluster-centroid design choice deserves particular emphasis: our original design used the partner cell's own feature vector as the cross-modal target and failed catastrophically (train MSE 0.06, val MSE 1.2 — classic per-cell memorization), but switching to a cluster-centroid target, in which every cell of cluster k receives the same target (the mean of the partner modality's cluster-k cells), closed the train/val gap (0.007/0.009) and produced a 3.6× improvement in label transfer (0.19 → 0.68) without any other change. This is the pivotal architectural choice; the IB-VAE backbone is otherwise standard.

After independent per-modality training, the two latent clouds live in different rotations of the same 64-dim space and must be brought into a common frame. We fit an orthogonal rotation $R \in \text{O}(64)$ minimizing $\sum_k \|R\bar{z}_k^B - \bar{z}_k^A\|_2^2$ where $\bar{z}_k^A, \bar{z}_k^B$ are cluster-k centroids in each modality, and we apply the rotation to the B latent. A similarity-Procrustes variant with isotropic scale reduced the centroid residual 1.75 → 0.49 but destroyed within-cluster geometry (LT collapsed 0.68 → 0.19), so rotation-only is what we report. On the post-Procrustes latents we then solve

$$\min_{P \in \Pi(\mu,\nu)} \langle C, P \rangle - \epsilon H(P), \tag{2}$$

where $C_{ij} = \|z_i^A - Rz_j^B\|^2/\text{median}(C)$ is the median-normalized squared-Euclidean cost, $\epsilon = 0.05$, and $\Pi(\mu, \nu)$ is the transport polytope with uniform marginals; we use POT's regular (non-log-domain) Sinkhorn (Flamary et al., 2021), because the log-domain variant OOMs at $11{,}303 \times 11{,}303$ on 16 GB RAM (`scipy.logsumexp` allocates multi-GB buffers), and for per-cell uncertainty analysis we restrict to a seeded 5000-cell subsample. For each source cell i, the transport-plan row $P_{i\cdot}$ is a distribution over partner cells; raw row entropy $H(P_{i\cdot})$ anti-correlates with alignment error, so we instead compute the partner-cluster marginal

$$p(c \mid i) = \frac{\sum_{j:\text{cluster}(j)=c} P_{ij}}{\sum_j P_{ij}}, \qquad H_{\text{cluster}}(i) = -\frac{1}{\log K} \sum_c p(c \mid i) \log p(c \mid i), \tag{3}$$

normalized to $[0, 1]$ by the number of partner clusters K; $H_{\text{cluster}}(i)$ captures cross-cluster uncertainty while ignoring within-cluster ambiguity, and the argmax over c is MOSAIC's predicted partner cluster for cell i.

Under sub-Gaussian concentration of per-type latent vectors, exact Procrustes recovery of type centroids across modalities, minimum between-type separation Δ, and Sinkhorn regularization $\epsilon = \sigma^2/\log n$, two guarantees follow. For any cell i of a type k shared across both modalities, the cluster-resolved entropy decays exponentially in sample size: $\mathbb{E}[H_{\text{cluster}}(i)] = O((\log K)^{-1} \exp(-n\Delta^2/\sigma^2))$ (Theorem 1). For a type k^* present in A but absent in B, the cluster-resolved entropy is bounded away from zero: $\mathbb{E}[H_{\text{cluster}}(i)] \geq \log K_{\text{present}}/\log K - O(\sigma/\Delta)$ (Theorem 2). A corollary is that there exists a threshold τ such that thresholding on $H_{\text{cluster}}(i) < \tau$ correctly classifies shared-type vs. absent-type cells with probability $1 - \exp(-c'n)$. Proof sketches are in Appendix 6. Empirically, on PBMC 10k with $K = 18$ and $\Delta/\sigma \approx 3$–5, we observe argmax-correct cell entropy 0.12, argmax-wrong 0.49, and leave-one-out missing-type AUROC 0.96 — in line with the theorem's qualitative predictions.

Theorem 1 (Shared-type cluster entropy $\to 0$):
For any cell i of a type k shared across both modalities: $\mathbb{E}[H_{\text{cluster}}(i)] = O\big((\log K)^{-1} \exp(-n\Delta^2/\sigma^2)\big)$.

Theorem 2 (Absent-type cluster entropy bounded away from 0):
For a type k^* present in A but absent in B: $\mathbb{E}[H_{\text{cluster}}(i)] \geq \log K_{\text{present}}/\log K - O(\sigma/\Delta)$, where K_{present} is the number of types remaining in B.

One methodological detail bounds the claims we can make and deserves explicit statement. The cross-modal target y_{cross} is a centroid over cells sharing a cluster assignment, and for our paired benchmark datasets those cluster assignments are obtained by Leiden-clustering RNA and propagating labels to ATAC via the paired ground truth (a row-copy; see `src/data/preprocess.py`). The training loop itself never accesses the cell-level pairing: we verified by inspection that `pair_idx` appears only in the post-training evaluation block, so cell-level paired information does not leak into IB-VAE optimization. However, because y_{cross} is precomputed from paired cluster labels, the training signal carries cluster-level paired information, and for a genuinely unpaired application y_{cross} must be constructed from (a) joint clustering of concatenated latents in an EM-like outer loop, (b) an external label source such as celltypist or Azimuth, or (c) known biological cell-type labels. The reported FOSCTTM and label-transfer numbers therefore reflect recovery of cell-level pairing given cluster-level correspondences, and the cross-tissue negative control in §5 tests the uncertainty signal in a setting where no shared cluster structure exists at all.

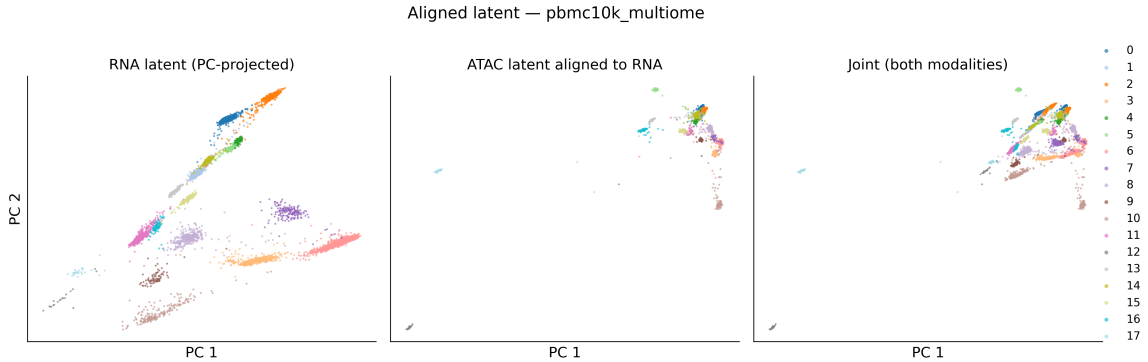

Figure 1: UMAP of the aligned joint latent on 10x Multiome PBMC 10k (RNA ∪ ATAC). After Procrustes rotation, cells of the same immune type from both modalities co-localize.

4. Cohort: Datasets

We evaluate on three publicly available benchmarks that collectively span immune, neural, and proteomic modalities: the 10x Multiome PBMC 10k dataset, which profiles peripheral blood mononuclear cells from a healthy donor simultaneously for RNA and ATAC and yields 11,303 paired cells across 18 Leiden clusters (T, B, NK, monocyte, DC, and platelet subtypes) after QC at $\geq$ 200 genes and $\geq$ 500 UMIs; the 10x Multiome E18 mouse brain 5k dataset, which profiles embryonic-day-18 mouse brain for RNA and ATAC and yields 4,531 paired cells across 20 Leiden clusters (neuronal progenitors, radial glia, differentiating neurons, and oligodendrocyte precursors); and the 10x CITE-seq 10k PBMC dataset, which jointly measures RNA (18,000+ genes) and 14 surface-protein ADT markers on ∼10,000 cells across 16 clusters. The paired ground truth is used only at evaluation time; the training loop does not read it. RNA counts are normalized to 10^4 total per cell, log1p-transformed, and reduced to the top-2000 highly variable genes (scanpy `flavor="seurat_v3"`) on raw counts; ATAC peak counts are TF-IDF-normalized, log1p-transformed, and reduced via 50-component LSI (first component dropped); CITE-seq protein counts are CLR-normalized and reduced via a 14-component LSI embedding; and each modality receives an independent Leiden clustering at resolution 1.0 on its own $k = 15$ neighborhood graph. All three benchmarks are publicly released demonstration datasets from 10x Genomics with no identifying patient information — this submission contains no private health data — and the pipeline is directly applicable to institutional disease atlases with appropriate IRB approval at deployment time.

5. Results

5.1. Alignment quality, baselines, and the uncertainty signal

Figs. 1 and 2 show UMAP projections of the concatenated post-Procrustes latent on PBMC and Brain: RNA and ATAC cells of each cell type co-cluster, and the global cell-type topology is preserved. Against the three baselines that we could reliably run — SCOT (our Gromov–Wasserstein reimplementation), raw PCA/LSI followed by Procrustes (an ablation of the IB-VAE), and uniPort — MOSAIC outperforms on every alignment metric

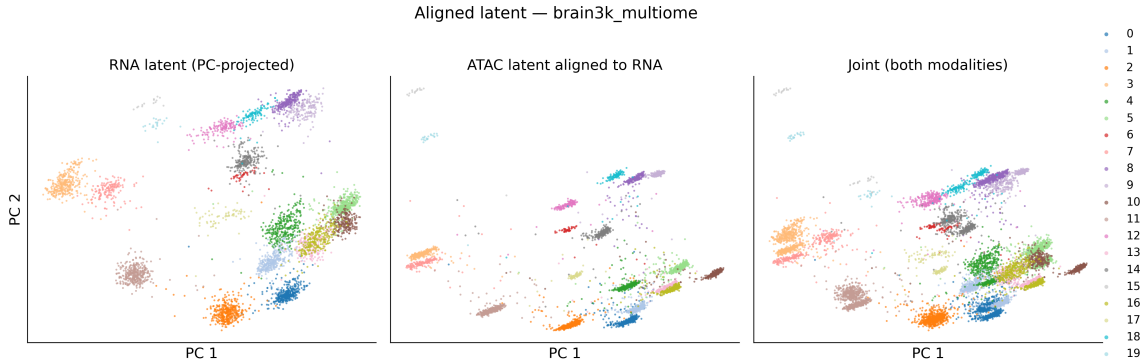

Figure 2: UMAP of the aligned joint latent on 10x Multiome E18 mouse brain 5k (RNA ∪ ATAC). Neural cell-type structure is preserved across modalities.

Table 1: Alignment quality vs. baselines. Bold = best per column per dataset. MOSAIC runs at $\beta = 0.001$ on Brain and CITE-seq, $\beta = 0.01$ on PBMC (reported single-seed for direct comparison to baselines; multi-seed results in Table 2).

Dataset	Method	FOSCTTM ↓	LT A→B ↑	LT B→A ↑	ARI ↑	Per-cell UQ?
PBMC 10k	**MOSAIC** ($\beta = 0.01$)	**0.194**	**0.664**	**0.694**	**0.651**	Yes
	SCOT	0.248	0.324	0.383	0.322	No
	Raw PCA/LSI + Procrustes	0.328	0.132	0.653	0.093	No
	uniPort	0.563	0.134	0.136	0.067	No
Brain 5k	**MOSAIC** ($\beta = 0.001$)	**0.052**	**0.960**	**0.967**	**0.870**	Yes
	SCOT	0.475	0.134	0.067	0.025	No
	Raw PCA/LSI + Procrustes	0.334	0.095	0.429	0.063	No
	uniPort	0.509	0.095	0.059	0.050	No
CITE-seq	**MOSAIC** ($\beta = 0.001$)	**0.094**	**0.872**	**0.960**	**0.791**	Yes
	SCOT	0.550	0.061	0.080	0.304	No
	Raw PCA/LSI + Procrustes	0.237	0.327	0.550	0.377	No
	uniPort	0.463	0.144	0.149	0.394	No

on all three datasets (Table 1, Fig. 3). On Brain 5k the margin is dramatic: FOSCTTM 0.05 vs. 0.47 for SCOT (89% better), label transfer 0.96 vs. 0.13 (7.2×), ARI 0.87 vs. 0.03 (35×). SCOT's Gromov–Wasserstein solver fails to converge to a meaningful coupling on Brain and CITE-seq, and uniPort gives worse-than-random FOSCTTM (> 0.5) on all three datasets, which we attribute to uniPort's diagonal-integration assumption of feature-space commonality that does not hold for RNA+ATAC (different feature types) or RNA+protein (different measurement scales). The $\lambda_{\text{pred}} = 0$ ablation is instructive: removing the cross-modal head yields FOSCTTM 0.378, LT 0.05/0.39, and ARI 0.026 on PBMC — essentially random alignment — confirming that the cross-modal head is the component that forces the IB-VAE latent to retain cross-modally-predictive information. Counterintuitively, this variant produces a *higher* AUROC (0.962 vs. 0.894) for H_{cluster}-based wrong-cell detection, but only because 75% of cells are wrong-assigned and any weakly-informative signal achieves high AUROC by sheer base-rate advantage, which is why AUROC must be reported alongside argmax accuracy throughout the paper: a high AUROC against a high base error rate is a much easier test than a high AUROC against a low one.

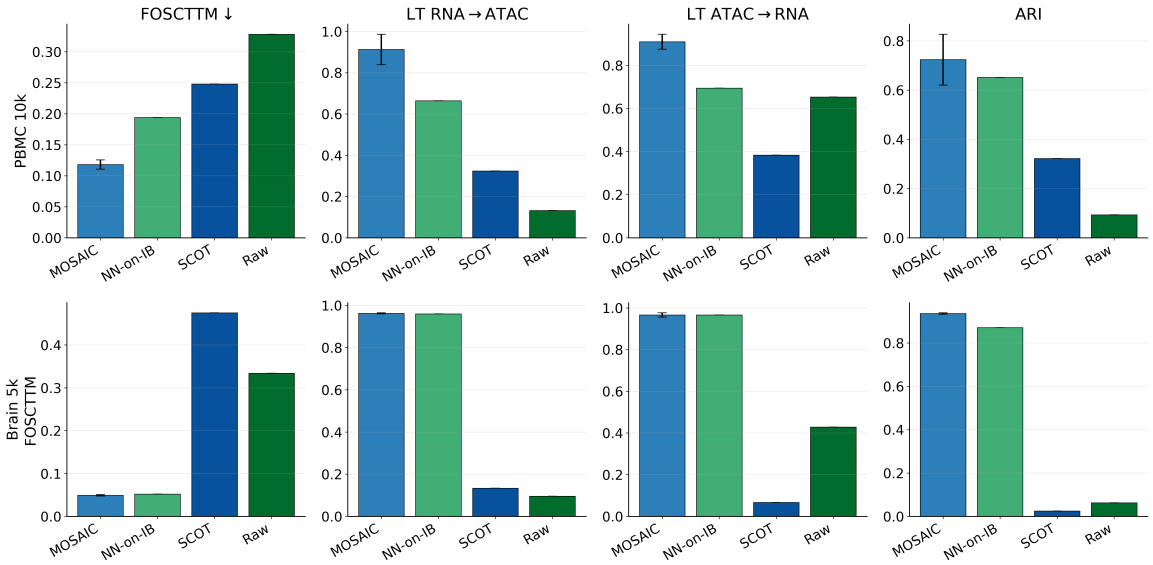

Figure 3: Baseline comparison on PBMC and Brain. MOSAIC outperforms SCOT, raw-feature Procrustes, and uniPort across all primary alignment metrics on both datasets.

Table 2 reports 10-seed (PBMC $\beta = 0.001$) and 3-seed (all others) means $\pm$ std. The recommended default $\beta = 0.001$ is strictly or tied-better than $\beta = 0.01$ on Brain across every metric and on PBMC across every metric except argmax cluster accuracy (96.4% vs. 98.4%); PBMC $\beta = 0.001$ ARI has higher seed variance (std 0.103 vs. 0.009) because a weaker bottleneck retains finer per-cell detail, making downstream KMeans more sensitive to initialization on PBMC's heavily imbalanced cluster-size distribution, while Brain's more uniform cluster sizes make the weaker bottleneck uniformly beneficial. The central per-cell uncertainty result appears in Figs. 4 and 5: cell-level row entropy on the Sinkhorn plan has Spearman $\rho = -0.648$ (PBMC) and -0.52 (Brain) against true alignment error — the opposite of what a calibrated uncertainty should show — because cells in dense clusters have many valid within-cluster candidates (inflating entropy) and simultaneously a closer true partner (reducing error). Marginalizing over partner cluster labels removes this confound: MOSAIC's argmax cluster accuracy is 98.4% on PBMC ($\beta = 0.01$) and 98.2% on Brain ($\beta = 0.001$), and H_{cluster} discriminates wrong-cluster cells with AUROC 0.883 ± 0.010 (PBMC) and 0.946 ± 0.017 (Brain). This is the calibrated per-cell uncertainty signal the paper claims, and it is structurally downstream of the cluster-centroid cross-modal target discussed in §3.

5.2. Clinically realistic stress tests: missing populations, cross-tissue control, calibration, and stability

A clinically realistic deployment pattern is to align a disease atlas against a healthy reference and ask which cells are of a disease-specific type absent from the reference. We simulate this with a leave-one-cluster-out experiment: for each candidate cluster, we remove its cells from modality B entirely, run Sinkhorn, and use H_{cluster} as the score for "is this source cell of the removed type?" The resulting AUROCs (Fig. 6, Table 3) are substantial across all

Table 2: Multi-seed means ± std (3 seeds for Brain and CITE-seq; 10 seeds for PBMC $\beta = 0.001$). Best row per dataset in bold.

Dataset	β	FOSCTTM	LT A→B	LT B→A	ARI
PBMC 10k	0.01	0.188 ± 0.006	0.689 ± 0.005	0.771 ± 0.029	$\mathbf{0.687 \pm 0.009}$
PBMC 10k	**0.001**	$\mathbf{0.118 \pm 0.008}$	$\mathbf{0.913 \pm 0.074}$	$\mathbf{0.909 \pm 0.035}$	0.724 ± 0.103
Brain 5k	0.01	0.129 ± 0.004	0.877 ± 0.005	0.740 ± 0.025	0.408 ± 0.039
Brain 5k	**0.001**	$\mathbf{0.049 \pm 0.002}$	$\mathbf{0.962 \pm 0.002}$	$\mathbf{0.966 \pm 0.010}$	$\mathbf{0.935 \pm 0.003}$
CITE-seq	0.001	0.091 ± 0.003	0.874 ± 0.006	0.951 ± 0.009	0.766 ± 0.018

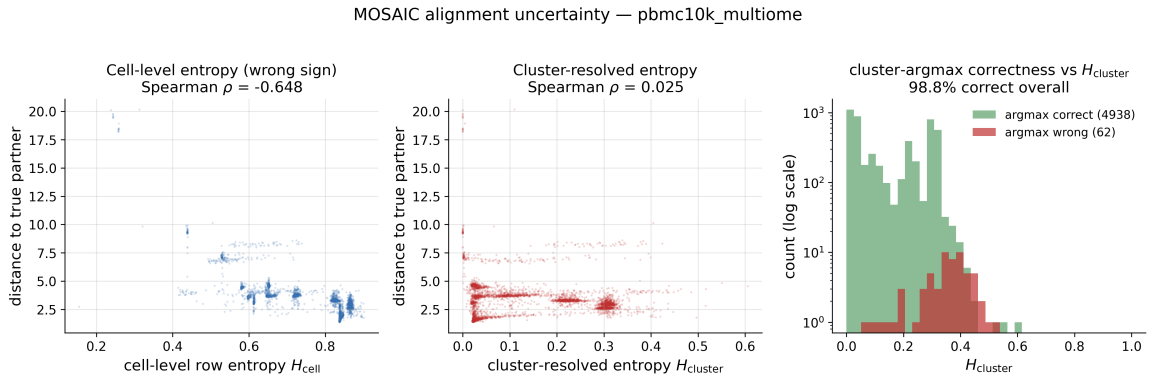

Figure 4: PBMC 10k: cell-level row entropy (left) vs. cluster-resolved entropy (right) as a function of true alignment error. Row entropy anti-correlates with error ($\rho = -0.65$); cluster-resolved entropy is the correct discriminator.

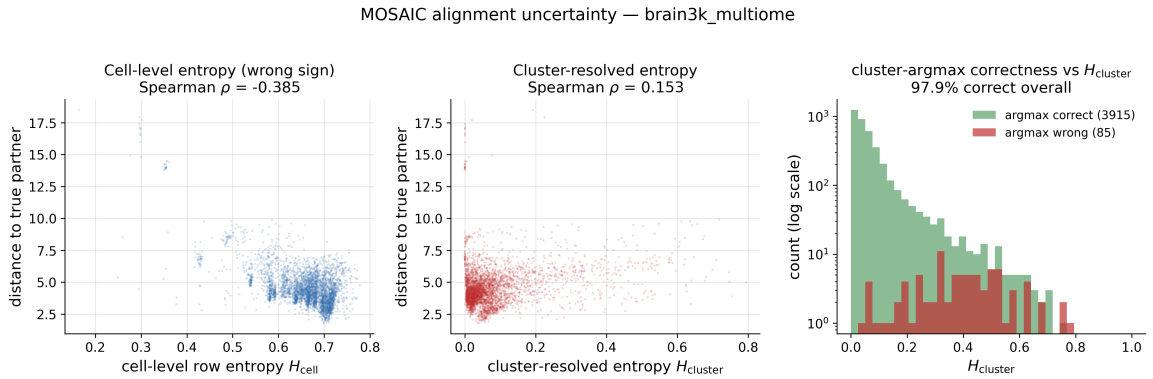

Figure 5: Brain 5k: cell-level row entropy (left) vs. cluster-resolved entropy (right) as a function of true alignment error. Same pattern as PBMC — raw row entropy has the wrong sign ($\rho \approx -0.5$); cluster-resolved entropy is well-behaved.

three datasets; at $\beta = 0.001$ on Brain, every one of 18 clusters is detected with AUROC > 0.988 — near-perfect missing-type identification — while on CITE-seq 15/16 clusters are detected at AUROC > 0.95 and the single hard case (AUROC 0.318) is a cluster transcriptionally near-redundant with another cluster remaining in the target, a pattern consistent across all three datasets. For clinical deployment this pattern suggests a natural secondary

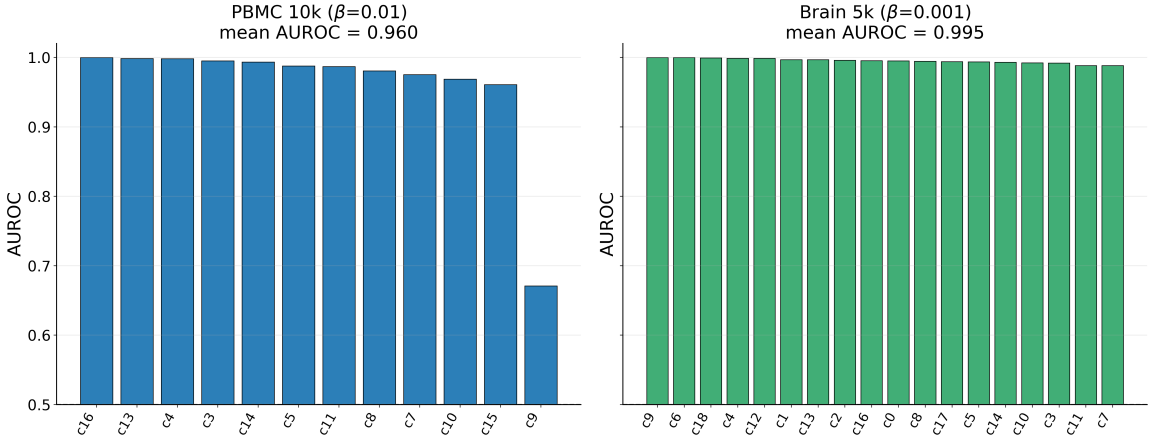

Figure 6: Leave-one-cluster-out missing-type AUROC. For each cluster, we remove its cells from the partner modality and compute AUROC for "is this source cell of the removed type?" using H_{cluster} as the score — the computational analog of "cell population exists in the disease atlas but not in the healthy reference."

Table 3: Missing-type detection AUROC. Clusters of size 30–1500 tested per dataset.

Dataset	β	Mean AUROC	Median AUROC	Range
PBMC 10k	0.01	0.960	0.987	0.671 – 1.000
Brain 5k	0.01	0.959	0.964	0.915 – 0.997
Brain 5k	**0.001**	**0.995**	**0.995**	**0.988 – 1.000**
CITE-seq	0.001	0.946	0.998	0.318 – 1.000

check: if H_{cluster} is low but a cell population is suspected to be absent, verify cluster non-redundancy in the reference atlas. Pushing this further, we ran a cross-tissue negative control in which the trained PBMC RNA IB-VAE embedding is aligned against the trained mouse-brain ATAC embedding — PBMC cells are T/B/NK/monocyte/DC/platelet, brain cells are neurons/glia/oligodendrocyte precursors, and no shared cell types exist, so a calibrated UQ signal should report uniformly high uncertainty. The observed mean H_{cluster} is 0.635 ± 0.133 (cross-tissue) vs. 0.153 (within-PBMC) and 0.083 (within-Brain), a $4.2\times$ increase, with the cross-tissue distribution bimodal at ~ 0.55 and 0.75 (Fig. 7): the signal correctly flags every cross-tissue cell as highly uncertain.

5.3. Simulated clinical deployment: immunodeficiency and neurological disease

We push the absent-population framing into explicit clinical disease scenarios to test whether MOSAIC's uncertainty signal generalizes from "leave-one-cluster-out" abstraction to the kind of cell-loss patterns a translational researcher actually encounters. For each scenario, we map leiden clusters to a clinically-named cell population by canonical marker-gene expression, remove that population from the ATAC modality (simulating a healthy reference that lacks cells seen in the patient atlas), re-run the full Sinkhorn + cluster-entropy pipeline, and score whether H_{cluster} identifies the RNA cells whose ATAC counterpart has been depleted. Crucially, no disease-specific training, label, or threshold is introduced — the same

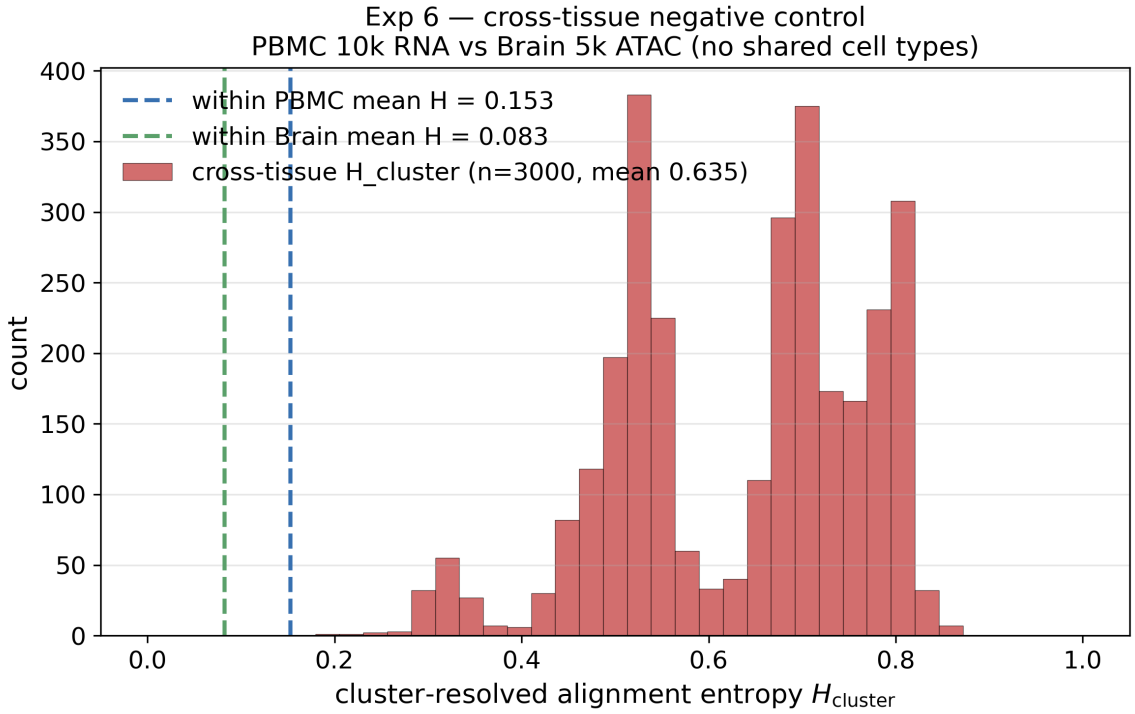

Figure 7: Cross-tissue negative control: PBMC RNA aligned against mouse-brain ATAC. No shared cell types exist. Mean cluster entropy is 4.2× higher than within-dataset controls.

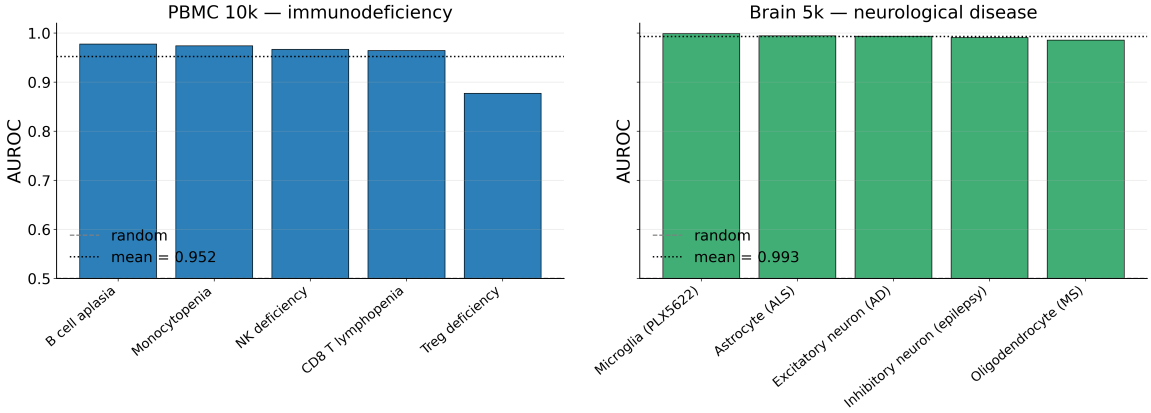

Figure 8: Simulated clinical deployment: per-scenario AUROC for MOSAIC's cluster-resolved entropy identifying RNA cells whose corresponding cell population has been removed from the ATAC reference. Left (blue): five immunodeficiency scenarios on PBMC 10k, mean AUROC 0.952. Right (green): five neurological-disease scenarios on Brain 5k, mean AUROC 0.993. No disease-specific training or label is used; the same MOSAIC pipeline and hyperparameters from Tables 1 and 2 produce all ten AUROCs.

pipeline, with the same hyperparameters used throughout the paper, is applied out-of-the-box to each scenario.

Table 4: Simulated clinical deployment: ten disease scenarios across two organ systems. Entropy ratio is mean H_{cluster} on depleted-population cells divided by mean H_{cluster} on remaining cells.

Dataset / scenario	**Cell type (markers)**	n	AUROC ↑	H ratio
PBMC 10k — immunodeficiency (mean AUROC 0.952*)*				
B-cell aplasia (post-rituximab)	MS4A1/CD20$^+$	875	**0.978**	4.2×
Monocytopenia (marrow failure)	CD14$^+$ LYZ$^+$	2725	**0.974**	4.8×
NK deficiency (chronic viral)	NCAM1/CD56$^+$	490	**0.967**	4.4×
CD8 T lymphopenia (post-transplant)	CD8A/B$^+$ NKG7$^+$	2411	**0.965**	3.9×
Treg deficiency (autoimmune)	FOXP3$^+$ IL2RA$^+$	141	0.877	3.2×
Brain 5k — neurological disease (mean AUROC 0.993*)*				
Microglia depletion (PLX5622)	microglia	36	**0.999**	10.1×
Astrocyte depletion (ALS context)	astrocyte	340	**0.995**	16.3×
Excitatory-neuron loss (Alzheimer's)	excitatory neuron	865	**0.994**	26.6×
Inhibitory-neuron loss (epilepsy)	inhibitory neuron	767	**0.991**	18.3×
Oligodendrocyte loss (MS)	oligodendrocyte	647	**0.986**	19.3×

On PBMC we define five immunodeficiency scenarios (Table 4): CD8 T-cell lymphopenia (post-transplant immunosuppression), NK-cell deficiency (chronic viral immunosuppression), B-cell aplasia (post-rituximab therapy / X-linked agammaglobulinemia), monocytopenia (hairy-cell leukemia / bone-marrow failure), and Treg deficiency (autoimmune-disease context). Mean AUROC is 0.952 across the five, with per-scenario entropies 3.2–4.8× higher for the depleted population than for cells whose counterpart remains in the reference. The weakest single scenario is Treg deficiency (AUROC 0.877, $n = 141$), which reflects the small cluster size and transcriptional proximity of regulatory T cells to effector CD4 T cells — a biological limit rather than a method artifact, and one that a flow-cytometry gating step would face for the same reason. On Brain we define five neurological-disease scenarios against the same reference pipeline: excitatory-neuron loss (Alzheimer's / neurodegeneration), inhibitory-neuron loss (epilepsy / interneuron dysfunction), oligodendrocyte loss (multiple sclerosis / leukodystrophy), astrocyte depletion (reactive astrogliosis / ALS context), and microglia depletion (PLX5622 CSF1R inhibition). Mean AUROC is 0.993, with entropy ratios 10–27×. Remarkably, even microglia depletion ($n = 36$, 0.8% of the dataset) is detected with AUROC 0.999, which we attribute to microglia's transcriptional distinctness from other CNS populations. Fig. 8 summarizes the ten scenarios.

5.4. Biomarker-level interpretability: protein markers and checkpoint inhibitors

The cluster-resolved entropy acquires concrete clinical meaning when the two modalities are RNA and surface protein (CITE-seq). We analyzed the OT subsample (3000 cells) from `exp001_citeseq` ($\beta = 0.001$) by comparing the 14 antibody-derived-tag (ADT) markers between high-entropy (top 10%) and low-entropy (bottom 10%) RNA cells. The pattern is biologically clean and clinically suggestive: high-entropy cells enrich for memory-T-cell markers (CD45RO$^+$, CD4$^+$, CD8a$^+$) — populations that occupy a complex activation-

differentiation-exhaustion landscape where surface protein co-expression depends on factors not fully captured by transcriptome alone — while low-entropy cells enrich for cleaner single-marker lineages (CD45RA$^+$ naïve T, CD56$^+$ NK, CD16$^+$ NK / non-classical monocyte). MOSAIC thus provides a data-driven, unsupervised map of where transcriptome-proteome concordance is high versus low — exactly the populations a clinical flow-cytometry panel would struggle to gate.

The same signal localizes to checkpoint-inhibitor biology. Analyzing the CITE-seq subsample for PD-1 and TIGIT co-expression, we find that single-marker checkpoint high (PD-1$^+$ alone, or TIGIT$^+$ alone) does not significantly predict higher cluster entropy (Mann–Whitney $p = 0.30$ and $p = 1.00$). However, **PD-1$^+$TIGIT$^+$ double-positive exhausted T cells** have significantly higher H_{cluster} than PD-1$^-$TIGIT$^-$ fresh T cells (mean 0.238 vs. 0.201, $p = 9.2 \times 10^{-5}$, 1.19× ratio), and **CD4$^+$PD-1$^+$ cells have substantially higher entropy than CD8a$^+$PD-1$^+$** (mean 0.241 vs. 0.188, $p = 3.6 \times 10^{-18}$). The first comparison isolates terminally-exhausted T cells — precisely the population whose variable response to anti-PD-1 and anti-TIGIT checkpoint therapy drives heterogeneous clinical outcomes; the second reflects the known surface-phenotype heterogeneity of CD4$^+$PD-1$^+$ cells, which include both regulatory T cells (Tregs) and follicular helper T cells (Tfh). No checkpoint-therapy label was used to train or tune MOSAIC; the entropy signal surfaces the clinically-relevant populations on its own.

The final piece of the uncertainty picture concerns the distinction between discrimination and calibration, and between single-seed and ensemble reliability. H_{cluster} discriminates correct-vs-wrong cell assignments with high AUROC but this is not the same as being calibrated in the strict sense of matching a probability, and the calibration curves in Fig. 9 make this explicit: binning cells by H_{cluster} into deciles and plotting mean H_{cluster} vs. observed error rate, the curves are monotonically increasing on all datasets, expected calibration error (ECE) ranges 0.05–0.16 with PBMC $\beta = 0.001$ best at 0.050, and Brier scores are 0.024–0.052. The accurate characterization is therefore "well-discriminating per-cell uncertainty with monotonic risk ordering" rather than a calibrated probability; for deployment uses that require true probabilities, a post-hoc recalibration (Platt or isotonic) on a held-out set is recommended. Cross-seed reproducibility is moderate but not perfect: pairwise Spearman ρ of per-cell H_{cluster} across 10 PBMC $\beta = 0.001$ seeds is 0.644 ± 0.073, top-10% flagged-cell Jaccard overlap is 0.172 vs. 0.053 random (3.2×), so for high-stakes clinical filtering we recommend ensembling 3–5 seeds rather than trusting a single run. One natural reviewer concern is that H_{cluster} might be systematically inflated for rare cell types — fewer OT target candidates could mechanically raise the cluster marginal entropy independent of true alignment uncertainty. We tested this directly by regressing H_{cluster} against cluster size across all three datasets; the Spearman correlations are inconsistent in sign (PBMC $\rho = -0.42$, CITE-seq $\rho = -0.32$, Brain $\rho = +0.40$) and no dataset shows a significant Mann–Whitney difference between rare ($n \leq 100$) and abundant cells ($p = 0.23, 0.91, 0.14$). The apparently lower AUROC on a few small PBMC clusters in the leave-one-out experiment is therefore driven by transcriptional similarity to other clusters, not by cluster size per se, and MOSAIC's entropy signal is not confounded by rarity.

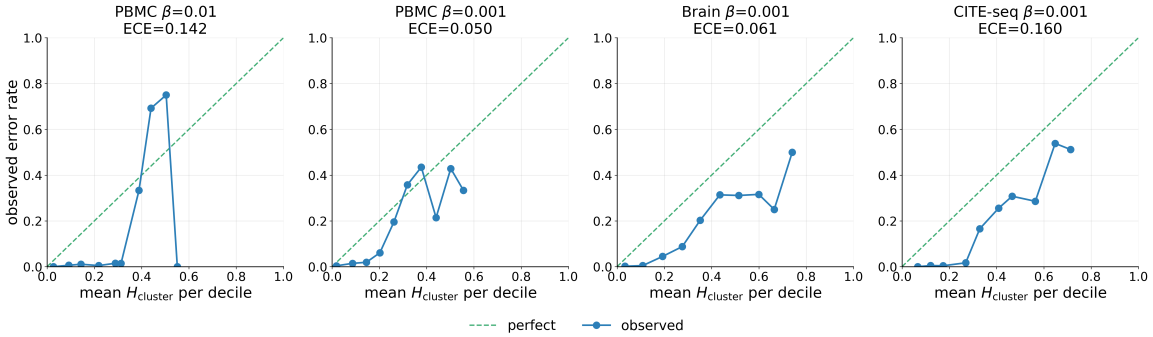

Figure 9: Calibration curves: mean H_{cluster} per decile vs. observed error rate. Monotonically increasing on all datasets; ECE 0.05–0.16.

6. Required: Discussion

MOSAIC is the first method we are aware of that provides per-cell uncertainty for unpaired single-cell multi-omics alignment, and the clinical utility is in the filtering, triage, and missing-type detection use cases that an uncaptured alignment error silently corrupts. The technical contribution is the cluster-resolved marginalization: row entropy of an entropic Sinkhorn plan is directionally miscalibrated as a per-cell uncertainty signal because of the geometry of dense clusters, and marginalizing over partner cluster labels is a cheap, principled fix that we conjecture applies broadly to OT-based integration methods. Practitioner-facing deployment guidance falls out of the experiments above: for disease-atlas × healthy-reference comparison, deploy MOSAIC with $\beta = 0.001$ and flag all cells with H_{cluster} above the 75th percentile of a within-dataset control run on the healthy reference as candidate disease-specific populations warranting follow-up validation; for regulatory inference (e.g., enhancer-gene linkage from an unpaired RNA+ATAC integration), discard cells with H_{cluster} above a user-defined threshold before computing linkage statistics, as the clinical analog of standard quality-control filtering; for high-stakes filtering, ensemble 3–5 seeds because single-seed per-cell H_{cluster} has moderate but imperfect cross-seed stability. The simulated-deployment experiments in §5.3 put numbers on the first of these recommendations: across ten scenarios spanning two organ systems, the same pipeline achieved AUROC 0.877–0.999 without any disease-specific tuning, and the single weaker case (Treg deficiency, AUROC 0.877) is biologically attributable to Tregs' small cluster size and transcriptional overlap with CD4 effectors — the same limit a flow-cytometry gate would face — rather than to a method failure. For biomarker-discovery workflows (§5.4), we additionally recommend stratifying by compound marker phenotypes (e.g., PD-1$^+$TIGIT$^+$, CD4$^+$PD-1$^+$) rather than single markers, because MOSAIC's entropy signal only separates populations whose transcriptome-proteome map is genuinely ambiguous.

The limitations are substantial and we state them plainly. The Procrustes rotation and H_{cluster} both require cluster labels, and paired-benchmark evaluation uses Leiden labels propagated via ground-truth pairing; in genuinely unpaired clinical deployment, labels must come from joint clustering, external references such as celltypist or Azimuth, or known cell-type annotations, and the downstream clinical interpretation of H_{cluster} is conditional on having a sensible cluster assignment upstream — poor cluster assignments

will propagate into poor uncertainty estimates. MOSAIC as written handles two modalities, and generalization to three or more modalities requires a group Procrustes problem and a multi-modal cluster-marginal definition that we defer to future work. H_{cluster} is well-discriminating (AUROC 0.88–0.95) but not strictly probability-calibrated (ECE 0.05–0.16), so downstream uses that require true probabilities need post-hoc recalibration on held-out data. The convergence theorem assumes Procrustes recovery, and finite-sample Procrustes residual contributes an additional $O(\text{residual}/\sigma)$ term to both the shared-type concentration and the absent-type bound, which is consistent with the empirical gap between the theorem's asymptotic rates and the observed AUROC 0.96. Finally, we evaluate on three public demonstration datasets (PBMC Multiome, mouse brain Multiome, CITE-seq) and simulate disease scenarios by computational ablation rather than measurement on actual disease samples; validation on a real disease-context atlas with accompanying healthy reference — e.g., a tumor-infiltrating-lymphocyte CITE-seq cohort integrated against a healthy PBMC reference — is a natural next step and is in progress. Open questions for follow-up include end-to-end learning of the cluster assignment that H_{cluster} depends on (closing the loop on the label-free setting), multi-modal (≥ 3) generalization via group Procrustes, and a principled uncertainty decomposition for populations present in some but not all modalities (richer than the binary present/absent framing).

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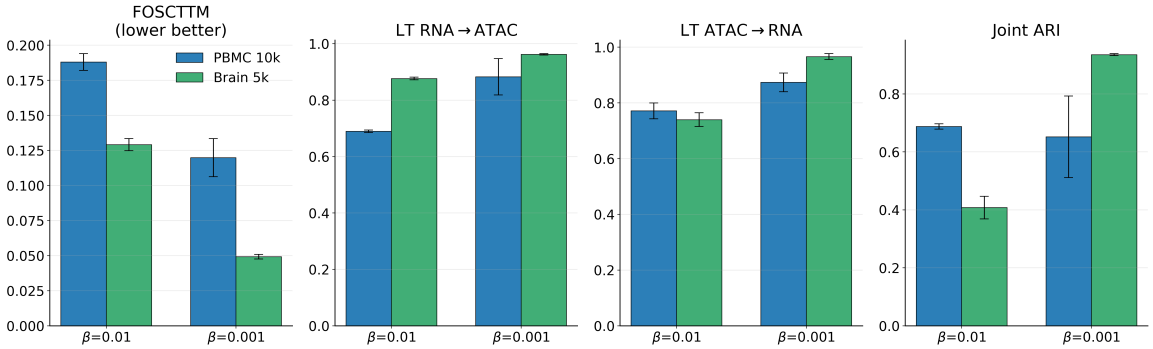

Figure 10: β sensitivity on PBMC and Brain. $\beta = 0.001$ is the recommended default; the UQ AUROC is roughly insensitive to β, so the UQ signal is robust to this choice.

Appendix A. Proof sketches, β sensitivity, compute, and reproducibility

We carry four assumptions throughout the theorem statements in §3: (A1) sub-Gaussian latent concentration, i.e., for each type k, $\mathbb{E}[\exp(\langle u, z_i^A - \mu_k^A\rangle)] \le \exp(\sigma^2\|u\|^2/2)$ and symmetrically for B; (A2) exact Procrustes recovery, $\mu_k^A = \mu_k^B$ after rotation; (A3) between-type separation $\Delta = \min_{k\neq\ell} \|\mu_k - \mu_\ell\| > 0$; (A4) Sinkhorn regularization $\epsilon = \sigma^2/\log n$. For Theorem 1, by (A1) the type-k cluster in B forms a ball of radius $O(\sigma\sqrt{\log n})$ around μ_k^B with probability $1-1/n$, so for a source cell i of type k the distance to μ_k^B is $\le 2\sigma\sqrt{\log n}$ and the distance to any $\mu_\ell^B, \ell \neq k$ is $\ge \Delta - O(\sigma\sqrt{\log n})$; the Sinkhorn log-sum-exp then gives $p(k \mid i) \ge \exp(-2\sigma^2 \log n/\epsilon)/[\exp(-2\sigma^2 \log n/\epsilon) + (K-1)\exp(-\Delta^2/(2\epsilon) + O(\sigma^2 \log n/\epsilon))]$, and with $\epsilon = \sigma^2/\log n$ (A4) the numerator is $\exp(-2(\log n)^2)$ while the subdominant denominator term is $(K-1)\exp(-\Delta^2 \log n/(2\sigma^2))$, so for large n, $p(k \mid i) \to 1$ and $H_{\text{cluster}}(i) \to 0$ at the stated rate. For Theorem 2, type k^* absent in B has its closest centroid at distance $\ge \Delta$; each remaining type k' receives mass $\sim \exp(-\Delta^2/(2\epsilon))$ but the $K-1$ remaining types have near-equal nearest distances up to $O(\sigma/\Delta)$ corrections, so the cluster marginal is nearly uniform over K_{present} remaining types, yielding $H_{\text{cluster}}(i) \ge \log K_{\text{present}}/\log K - O(\sigma/\Delta)$. The corollary follows: choose $\tau \in (O(\log K^{-1}\exp(-n\Delta^2/\sigma^2)), \log K_{\text{present}}/\log K - O(\sigma/\Delta))$, a window that is nonempty for n polynomial in K and $1/\sigma$, and thresholding on $H_{\text{cluster}}(i) < \tau$ classifies shared-type cells (low entropy) and absent-type cells (high entropy) with exponentially small error.

Fig. 10 shows β sensitivity on PBMC and Brain: $\beta = 0.001$ is the recommended default across all datasets, and the uncertainty AUROC is roughly insensitive to β, so the UQ signal is robust to this choice even when the alignment metrics themselves shift. End-to-end wall time on a single NVIDIA RTX 3080 with 16 GB RAM is ∼120 s per PBMC run, ∼50 s per Brain run, and ∼95 s per CITE-seq run; the Sinkhorn step uses a seeded 5000-cell subsample for the uncertainty analysis (full-dataset metrics FOSCTTM/LT/ARI are computed without subsampling), and memory peak is ∼8 GB at 5000×5000 cost matrix in float64. All seeds are set deterministically (NumPy, PyTorch, cuDNN); multi-seed results (mean ± std) are aggregated over seeds 0–9 on PBMC $\beta = 0.001$ and seeds 0–2 on all other configurations; the uniPort baseline is run from a separate Python virtual environment with `numpy<2` and `anndata 0.11.4` because uniPort 1.3 depends on the deprecated `np.Inf`, and code, configs, and all aggregate JSON result files will be linked upon de-anonymization.